

Finite-dimensional representation space in the unital singleton case

`MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton_amongNonUnital`

theorem

Applies the unital finite-dimensional-space theorem after checking the representation-interface conversion.

Statement

Let A be a nonzero unital complex C^* -algebra, let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ satisfy the singleton condition of R11. Then H is finite dimensional over $\mathbb{C}$.

Assumptions

The universal condition quantifies over every nonzero irreducible $*$ -representation of A ; representations presented without a unit equation are unital by irreducibility, as explained in R11.

Conclusion

The same representation space H has a finite complex basis.

Proof route

Convert the comparison condition for the same representation and apply the finite-dimensional-space theorem.

▼ Proof steps

1. Apply The all-representations singleton condition implies the unital singleton condition to π . For a unital nonzero irreducible $*$ -representation ρ , write $\rho_0(a) = \rho(a)$ for the same operator map viewed through the interface that does not require a unit equation. The assumed condition then supplies a unitary U with $U\pi(a) = \rho(a)U$ for every $a \in A$. Hence π satisfies the unital singleton condition.
2. Apply Finite dimension of the unital singleton representation space to the same nonzero ordered unital A , same separable H , and same π with that converted condition. Its conclusion is $\dim_{\mathbb{C}} H < \infty$. In that theorem the closed compact-preimage ideal becomes all of A , so $\pi(1) = I_H$ is compact and the compact-identity criterion supplies the finite dimension.

Main citations

- Exact theorem and proof
- The all-representations singleton condition implies the unital singleton condition
- Finite dimension of the unital singleton representation space

- The singleton condition quantified through the unit-free representation interface

Lean source signature (exact)

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theorem finiteDimensional_space_of_singleton_amongNonUnital
  [Nontrivial A] [TopologicalSpace.SeparableSpace H]
  (pi : Representation A H)
  (hsingle : IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi) :
  FiniteDimensional ℂ H
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[TopologicalSpace.SeparableSpace H]	The complete complex Hilbert representation space H is separable.
(pi : Representation A H)	The specified unital $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$.
(hsingle : IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi)	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
FiniteDimensional ℂ H	The conclusion concerns the whole original Hilbert space H . The declaration does not replace it by a different representation space.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton_amongNonUnital

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